Visible Food Surface Anomaly Classification

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***Abstract*—The paper aims to research the issue of classification of visible food surface anomalies using deep learning to classify fruit images as fresh and rotten fruits. Two deep learning methods were used: a custom convolutional neural network (CNN) method and an ImageNet-pretrained EfficientNetB0 model on a public dataset including apples, bananas, and oranges in six classes. The dataset included 13,599 images, including 2,698 images for a held-out test set. Classification accuracy, macro-averaged F1-score, and anomaly recall were evaluated for both models, focusing especially on wrongly classified rotten fruits as fresh fruits. A custom CNN reached 89.33% accuracy, 89.43% macro F1-score, and 95.96% anomaly recall; 62 false negatives were obtained for this approach. The imageNET pretrained EfficientNetB0 reached 99.89% accuracy, 99.90% macro F1-score, and 99.80% anomaly recall, resulting in three false negatives. Thus, the results prove that using transfer learning was more effective in the analyzed case. Yet, the evaluation was performed only for three fruit types using one dataset, and the probability of the existence of duplicated images in the test sets has not been evaluated. The method finds patterns that are linked to the labels of a certain dataset and cannot guarantee food safety.**

***Index Terms*—Deep Learning, Food Surface Anomaly Detection, Fruit Classification, Image Classification Convolutional Neural Networks, EfficientNetB0, Transfer Learning, Image Classification, Visible Food Surface Anomalies**

I. Introduction

The assessment of food quality is an important process in the food industry, particularly when the changes which can be seen in fruits during during storage and handling is identifiable. Classical methods of visual inspection usually rely on human perception, and there is always some level of variability with regard to experience of the inspector, environmental light, and the appearance of food products. Recent progress in artificial intelligence and computer vision allowed to provide new tools for classifying images of food products using automated techniques. Deep learning models such as Convolutional Neural Networks (CNNs) have shown capabilities of extracting such visual properties of images as colors, textures, shapes, and surface patterns that may help differentiate between various classes of fruits and visually labelled conditions. There might be difficulties with classification due to variation in appearance of fruits, backgrounds, and surfaces, but transfer learning allows to employ knowledge gained from big datasets of images in order to improve classification performance in more specific cases. Thus, comparison of classical CNN networks with pretrained deep learning models presents an opportunity to evaluate their efficiency at detecting surface anomalies of foods.

Visible surface anomalies that are characteristic of fruits are applicable to image classification for purposes of automation of visual screening and quality assessment. Changes in colour, texture, and surface features can occur when fruits spoil; however, only visible features cannot be sufficient to determine if a food product is safe to consume. Development of an image classification system that is able to classify those patterns will allow further development of the field of computer-assisted visual inspection. In this study, classification of images of fresh and rotten apples, bananas, and oranges from the publicly available dataset called Fresh and Rotten Fruits Classification will be considered. The two deep learning approaches will be tested in order to classify those images: a custom Convolutional Neural Network as a baseline and EfficientNetB0 architecture with transfer learning. The two models will be analyzed according to the metrics such as classification accuracy, macro F1-score, anomaly recall, and confusion matrices in order to compare their predictive performance. False negative predictions in particular (i.e., images that are labeled as rotten but classified as fresh) are of specific interest because they reflect missed anomalies.

II. Related Work

The use of deep learning has been found to be very relevant to the process of image analysis, especially when it comes to the identification and classification of objects in digital images. CNNs have been found to be used widely in image classification due to the ability to learn patterns automatically from training images like color, texture, shape, and spatial arrangement. The study conducted by Mureșan and Oltean (2018), titled Fruit Recognition from Images Using Deep Learning, focused on exploring the possibility of applying deep learning to recognize different fruit classes in images. While CNNs have been shown to be capable of learning useful representations from training images for automatic fruit recognition, unlike some conventional methods of image processing which require manual extraction of features, the recognition of different fruits is not the same as recognizing fresh or rotten fruits based on visual difference. This is because while classifying the fruits based on freshness, one will need to recognize differences in the appearance of surface of the fruit. This study has been built on the concept of image classification by focusing on the classification of six visually labeled classes of fresh and rotten apples, bananas, and oranges. Related research has also explored visual product-surface inspection

and food anomaly detection using visible, hyperspectral, and multimodal imaging [14]–[20].

The use of transfer learning is another popularly researched technique in the field of deep learning. Rather than learning all visual representations from the scratch, transfer learning enables models to reuse the knowledge gained from previously learned neural networks. Rather than learning all visual representations from the scratch, a pretrained model can be adapted for classification on a different dataset. In EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks by Tan and Le (2019), the authors proposed the family of EfficientNet architectures to create a better balance between classification performance and efficiency. In particular, the use of compound scaling is proposed for systematic scaling of network depth, width, and input image resolution. The EfficientNetB0 is the baseline architecture from the family of EfficientNets and can serve as the pretrained feature extractor for image classification tasks. As opposed to a custom CNN that needs to learn visual representations from scratch, the pretrained model of EfficientNetB0 starts with the visual representation of a large set of images like the ImageNet dataset. Thus, it creates an excellent ground for comparison with regards to the classification performance on a particular dataset. Hence, this paper compares the custom CNN with the transfer learning of EfficientNetB0. Other established CNN architectures include residual networks, densely connected networks, Inception, and MobileNets [4]–[7]; EfficientNetV2 further explores model size and training speed [3]. The use of ImageNet-pretrained representations is grounded in the scale of the ImageNet image database [8], while transfer learning has been widely surveyed as a strategy for adapting learned representations to new tasks [10].

The assessment of image classification models is a crucial part of deep learning studies because the overall accuracy does not always give a full picture of the model performance. The accuracy of classification is the fraction of the correctly classified images, but it does not reveal possible discrepancies in classification performance for each category separately. Precision, recall, and F1-score metrics are additional tools to evaluate the model performance. In particular, the macro F1-score is very helpful in multiclass classification problems as it gives the average F1-score across all classes with equal importance of each one. The confusion matrices provide additional information by revealing which classes are correctly classified and which are mistaken. In visible food surface anomaly classification, false-negative predictions are particularly important because they refer to the images classified as fresh although their labels indicate that they are rotten. Besides, the credibility of the experiment results is based on the proper division of training and test datasets because the presence of duplicated or similar images in both of them gives a biased estimate of the performance. Considering these evaluation challenges, the present study examines accuracy, Macro F1-score, anomaly recall, confusion matrices, and representative false-negative predictions to compare the performance of the custom CNN and EfficientNetB0 models. Precision, recall, and F-measure are established evaluation measures for classifier performance [11].

III. Dataset Description

A. Dataset Source

The data used in this paper was the Fruits Fresh and Rotten for Classification dataset which can be found on Kaggle [1]. In this dataset, images of fruits which belong to either of the two states -fresh or rotten based on the visual appearance- are labelled. There are three common types of fruit that are used. They include apples, bananas, and oranges. Each type of fruit is included in both fresh and rotten states for classification. These images have visual properties like colours, textures, shape, and pattern, which give information about the differences between the two categories. The dataset was chosen because it was aligned with the objective of creating an automated system for visible food surface anomalies using deep learning.

B. Dataset Classes

The dataset contains six different classes of images, corresponding to three fruit types in two different states of visualization. These classes are freshapples, freshbanana, freshoranges, rottenapples, rottenbanana, and rottenoranges. The first three classes contain images of fruit that have been labeled as fresh, while the last three classes have images of fruit labeled as rotten. The class represents the combination of fruit and the state of labeling, as shown in Table I. The task of classification involves the distinction of all six classes and not just the binary distinction between fresh and rotten fruit. Such an approach allows the models to learn the distinctive visual features of each fruit type along with those related to the fruit surface degradation. In terms of anomaly detection evaluation, the three classes of rotten fruit are considered to be the positive class of anomalies, while the fresh classes make up the negative one.

TABLE I
Fruit Categories Used in the Classification Task

Class Label	Fruit Type	Condition
freshapples	Apple	Fresh
freshbanana	Banana	Fresh
freshoranges	Orange	Fresh
rottenapples	Apple	Rotten
rottenbanana	Banana	Rotten
rottenoranges	Orange	Rotten

C. Dataset Distribution

The data set used in this experiment is made up of 13,599 images, which are divided into training, validation, and testing sets. The original training set consists of 10,901 images, whereas the testing set is comprised of 2,698 images. A validation split of 20% was performed on the original training set with a fixed random seed of 123. In this case, 8,721 images were assigned for model training, whereas 2,180 images were assigned for validation. The training set was used to learn

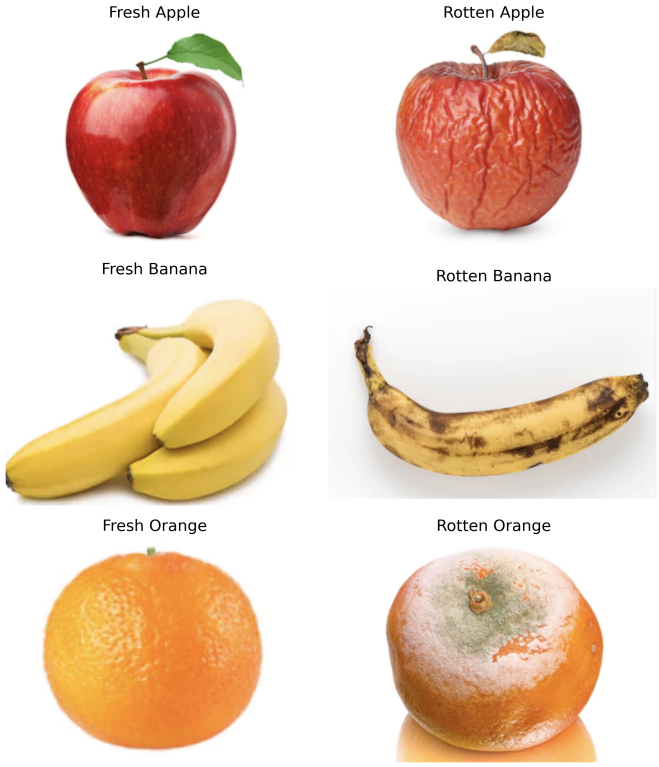


Fig. 1. Representative images of fresh and rotten apples, bananas, and oranges from the dataset.

the visual features from labeled fruit images. The validation set was used to monitor the performance of the model during training and test its generalization capabilities on new samples. The independent testing set was kept aside for the evaluation of the final classification performance of the trained models. All the input images were resized to 224×224 pixels and converted into three-channel RGB images with a batch size of 32.

TABLE II
DISTRIBUTION OF IMAGES ACROSS DATASET PARTITIONS

Partition	Images	Percentage
Training	8,721	64.13%
Validation	2,180	16.03%
Testing	2,698	19.84%
Total	13,599	100.00%

D. Sample Images

Examples of images taken from each of the six classes of datasets are shown in Fig. 1 to demonstrate the visual features of fresh and rotten fruits. The chosen samples consist of fresh and rotten apples, bananas, and oranges to cover all the classification categories. The fresh fruit images might show rather consistent colours, textures, and surfaces of fruit based on the fruit type. On the contrary, the images of rotten fruits might reveal some differences like changes in colours, dark spots, textured surfaces, and surface deterioration. These visual clues can be used for learning by convolutional neural networks while model training. However, since the appearance of fruit degradation varies between fruit types, the visual differences should be recognized by the models. Thus, the examples of images show that classification is done taking into consideration both fruit and degradation appearance. Hence, the chosen samples demonstrate the visual classification problem examined through Custom CNN and EfficientNetB0 architectures.

IV. PROPOSED METHODOLOGY

The suggested approach examines the classification of fruit surface conditions based on two deep learning models – a custom Convolutional Neural Network (CNN) and an efficientnetb0-based transfer learning model. Both architectures were used to classify images in six classes which represented fresh fruits and rotten fruit (apples, bananas, and oranges). The experimental process consisted of data preparation, image preprocessing, model design, training, predicting, and the evaluation of results. A custom CNN was built as a baseline architecture that would enable learning of visual properties from raw training images. As opposed to the latter, efficientnetb0 used pretrained representations of ImageNet to extract visual features. Training of both models was performed using the same splits of the dataset in order to compare their classification abilities. Evaluation included such aspects as classification accuracy, F1-score, anomaly recall, as well as analysis of wrongly classified images. Overall, the method was aimed at assessing the effectiveness of CNN and transfer learning for the problem of identification of fruit surface anomalies.

A. Data Preprocessing

The data preprocessing was carried out with the aim of preparing the images of fruit for further use in training and evaluation of classification models. Loading was done by applying the TensorFlow `image_dataset_from_directory` tool that takes image samples based on directory names (labels). Image samples were downsized to the same resolution size of 224×224 pixels to have consistent input images throughout the experiments. The processing was done in RGB mode, whereby three colour channels were considered and they held visual information about the fruit appearances. The integer encoded class labels were adopted for labelling the six classes in supervised model training. A batch size of 32 was adopted to process several image samples within each training round. The training set of fruit images were split into the training set and validation set with 20% validation split and the random seed set to 123.

B. Data Augmentation

Data augmentation was considered as a method for creating variations of the initial fruit images by performing controlled transformations. Data augmentation consisted of random flipping, random rotation, and random zooming procedures carried out with the help of TensorFlow and Keras. Random

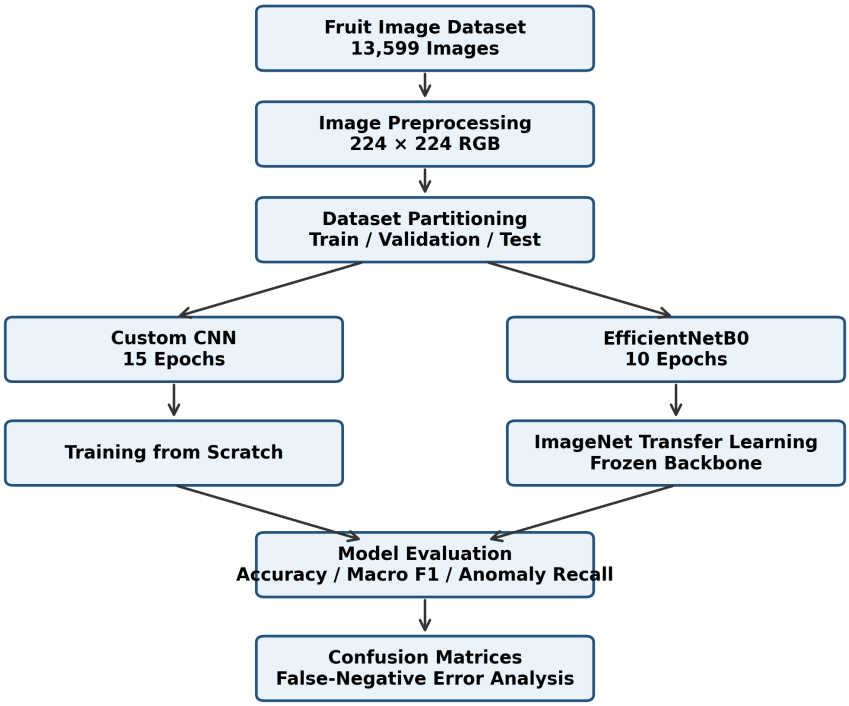


Fig. 2. Proposed methodology for visible food surface anomaly classification using Custom CNN and EfficientNetB0.

flipping alters the orientation of the image and creates different spatial representations of the same image content. Random rotation rotates fruit images but keeps their classes unchanged. Random zooming creates variations in the perceived size of the fruit in the image. Such transformations are common image data augmentation strategies [9]. All these transformations were illustrated to show how image variations can be created from the current dataset. Nevertheless, data augmentation was not incorporated into the training process of the custom CNN model and was applied only for illustration purposes. Therefore, the presented experimental results do not prove any performance improvement due to data augmentation.

C. Custom CNN Architecture

For the classification of fruit images in six classes, a Custom Convolutional Neural Network was employed as a basic neural architecture. This network comprised three convolutional layers with 32, 64, and 128 filters, respectively, in order to capture increasingly more complex representations. After each layer of convolution, max pooling was performed in order to reduce the size of dimensions and keep feature responses that were prominent. Each of these convolutional layers were constructed to detect the image attributes of fruits like the shapes of fruits, color differences, texture, and surface degradation. After that, the flattening process helped in turning multidimensional features into a one-dimensional feature vector. One fully connected layer having 128 neurons in it was used in order to establish the connection between the features of the image. In order to avoid overfitting, a dropout layer having a rate of 0.5 was employed. At last, an output layer comprising of six neurons with softmax activation generated probabilities of six fruit categories.

The custom CNN model was compiled with the use of the Adam optimizer [13] and the sparse categorical cross-entropy as the loss function. The Adam optimizer was used to update the parameters of the network during the training process. The sparse categorical cross-entropy loss was chosen due to the fact that the target classes were coded with integer-encoded class labels. The model was trained for 15 epochs using the prepared training set. The output layer of the network delivered the probability distribution over the six possible classes. The predicted class was selected based on the class label with the maximum output probability.

D. EfficientNetB0 Transfer Learning

The second architecture that used the approach of classification was EfficientNetB0. It consisted of convolutional features trained on the Imagenet dataset during pretraining of the model. Contrary to the previous model – custom CNN, EfficientNetB0 applied the already acquired feature representations. The pretrained version of EfficientNetB0 was loaded and modified so that it could be applicable to perform classification of six classes of fruits. EfficientNet uses compound scaling to balance network depth, width, and input resolution [**?**], and its

TABLE III
ARCHITECTURE OF THE CUSTOM CNN MODEL

Layer	Configuration
Input	$224 \times 224 \times 3$
Convolutional Layer 1	32 filters
Max Pooling Layer 1	Spatial downsampling
Convolutional Layer 2	64 filters
Max Pooling Layer 2	Spatial downsampling
Convolutional Layer 3	128 filters
Max Pooling Layer 3	Spatial downsampling
Flatten	Feature vector conversion
Fully Connected Layer	128 neurons
Dropout	0.5
Output Layer	6 neurons, Softmax

pretrained visual representations are based on ImageNet [8]. The parameters of the backbone were frozen to keep feature representations during the training phase. The global average pooling was applied in order to obtain a feature representation of the input image using the output feature maps of the backbone part of the neural network. Dropout layer with the parameter 0.3 and dense layer with 128 neurons were included to prevent overfitting within the classification branch.

Using the transfer learning method, the classification model could benefit from the visual representations created by the pretrained EfficientNetB0 architecture without fine-tuning the whole architecture. While training, the feature extractor network, which was frozen, provided the representation of the input images to the new classification layers. Global average pooling was performed on the extracted feature maps before the classification step. The dense layer worked on the feature representations to classify the input images into six categories of fruits. The softmax layer then produced the probability scores for fresh and rotten apples, bananas, and oranges. The EfficientNetB0 model was trained for 10 epochs with the training and validation datasets. The testing dataset was used to evaluate the performance of the model.

TABLE IV
EFFICIENTNETB0 TRANSFER LEARNING ARCHITECTURE

Component	Configuration
Input	$224 \times 224 \times 3$
Base Architecture	EfficientNetB0
Pretrained Weights	ImageNet
Original Classification Head	Excluded
Backbone Parameters	Frozen
Feature Pooling	Global Average Pooling
Dropout	0.3
Fully Connected Layer	128 neurons
Output Layer	6 neurons, Softmax
Training Epochs	10

V. EXPERIMENTAL SETUP

A. Training Environment and Configuration

The experiments were performed on Google Colab with TensorFlow and Keras as the tools to implement and train deep learning models. GPU was used to assist with calculations needed for image processing and neural network training. The custom CNN and EfficientNetB0 models were trained using images of RGB fruits resized to the resolution of 224×224 pixels. Batch size of 32 was used consistently throughout training to provide uniform image dataset processing. The Adam optimizer was chosen as the tool for parameter optimization [13] and sparse categorical cross-entropy was used as the loss function for classification into six classes. The custom CNN was trained from scratch for 15 epochs in order to learn visual features directly from the images of fruits. Meanwhile, EfficientNetB0 was trained for 10 epochs with pretrained ImageNet weights with frozen backbone layers. The configurations of the experiments were set to compare a regular CNN with the use of transfer learning.

TABLE V
EXPERIMENTAL TRAINING CONFIGURATION

Parameter	Custom CNN	EfficientNetB0
Input Image Size	224×224	224×224
Batch Size	32	32
Optimizer	Adam	Adam
Loss Function	Sparse CE	Sparse CE
Training Epochs	15	10
Pretrained Weights	None	ImageNet
Backbone Status	Trainable	Frozen
Output Classes	6	6

B. Evaluation Procedure

The evaluation process was meant to determine the performance of the custom CNN and EfficientNetB0 models in classifying the images in the same testing set. The initial training data consisted of 10,901 images, which were split into 8,721 training images and 2,180 validation images based on an 80:20 ratio. There was a testing set that had 2,698 images and was meant for testing the trained models. Both models classified the images into six classes, which were fresh and rotten apples, bananas, and oranges. Classification accuracy was computed as the percentage of images classified correctly in the testing set. Macro-averaged F1-score was used to test the performance of the two models in the six categories, considering that all the classes were equally important. Anomaly recall was determined by combining the rotten fruit classes as the positive anomaly class and the fresh fruit classes as the negative class.

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \tag{1}$$

$$F1_i = \frac{2P_iR_i}{P_i + R_i} \tag{2}$$

$$\text{Macro F1} = \frac{1}{K}\sum_{i=1}^{K} F1_i, \qquad K = 6 \tag{3}$$

$$\text{Anomaly Recall} = \frac{TP_{\text{rotten}}}{TP_{\text{rotten}} + FN_{\text{rotten}}} \tag{4}$$

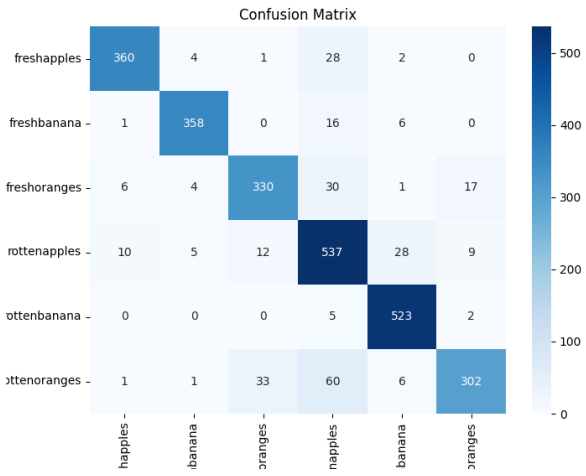


Fig. 3. Confusion matrix of the Custom CNN model evaluated on the six-class fruit test dataset.

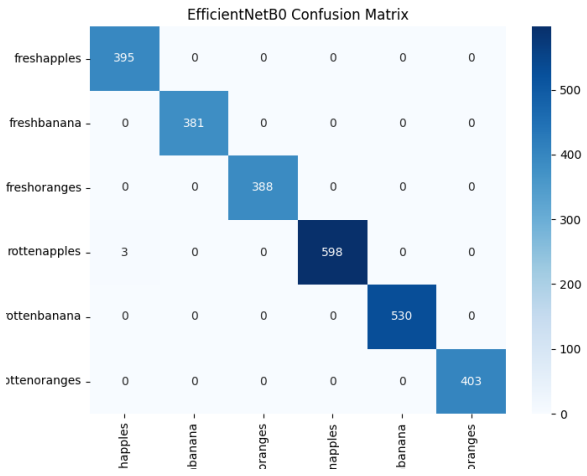


Fig. 4. Confusion matrix of EfficientNetB0 evaluated on 2,698 test images.

VI. Results and Evaluation

A. Model Performance Comparison

Both the custom CNN model and EfficientNetB0 were tested using the same test dataset, which had 2,698 images categorized into six fresh and rotten fruits. Custom CNN had a test accuracy score of 89.33while EfficientNetB0 scored an impressive accuracy of 99.89This indicates that there is an increase of 10.56 percentage points in classification accuracy scores. As for the macro-averaged F1-scores, there was a rise from 89.43in custom CNN to 99.90in EfficientNetB0, meaning that there is an increase in the efficiency of classification among the six categories. The custom CNN model had an anomaly recall of 95.96while EfficientNetB0 model had an anomaly recall of 99.80These results indicate that the use of EfficientNetB0 was more efficient in classifying rotten images in the testing dataset. The custom CNN wrongly classified 62 rotten images as fresh while EfficientNetB0 only made three errors in the process. In general, it can be noted that the use of transfer learning with EfficientNetB0 pre-trained features resulted in more efficient classification.

TABLE VI
Performance Comparison of Custom CNN and EfficientNetB0

Evaluation Metric	Custom CNN	EfficientNetB0
Test Accuracy (%)	89.33	99.89
Macro F1-score (%)	89.43	99.90
Anomaly Recall (%)	95.96	99.80
False Negatives	62	3

The performance of both the models on the basis of the selected metrics is summarized in Table VI. From the comparison, it can be seen that EfficientNetB0 outperformed the custom CNN model in terms of accuracy, macro F1-score, and anomaly recall. An improvement in macro F1-score implies that the transfer learning model performed better on average for all the fruit categories. An increase in anomaly recall implies that EfficientNetB0 could identify more images of the class 'rotten'. A decrease in the number of false negative predictions is significant because the focus of the project is to detect surface deterioration. The custom CNN showed that even a fairly basic CNN model could learn useful visual features from the training data. However, the pre-trained EfficientNetB0 architecture gave significantly better results on the same testing set.

B. Confusion Matrix Analysis

The classification behavior of the custom CNN and EfficientNetB0 models for the six fruit classes was analyzed using confusion matrices. The matrices show the relation between the actual class labels and predictions made by the respective models. Correct predictions are shown by numbers on the diagonal line, whereas incorrect predictions are shown by numbers that are not on the diagonal line. The custom CNN model made more classification errors, which is in line with the lower accuracy and macro F1-score compared to EfficientNetB0. EfficientNetB0 model made much fewer classification errors and had higher classification accuracy. The confusion matrices differentiate errors due to different fruit types from errors due to fresh and rotten states. Such differentiation is necessary since an error due to the wrong fruit type does not necessarily mean an inability to detect the anomaly in the image. Therefore, confusion matrix analysis was used in combination with anomaly recall.

Fig. **??** is the confusion matrix derived from the custom CNN model, whereas Fig. 4 shows the confusion matrix for EfficientNetB0. It can be observed that the classification accuracy of the custom CNN is somewhat lower than that of the EfficientNetB0, as there is a higher number of predictions off the main diagonal. The confusion matrix for EfficientNetB0 has significantly fewer predictions off the main diagonal, which is due to its higher classification accuracy. From this comparison, it can be concluded that the pretrained network is better in differentiating the visual features encoded into the

labels of the six classes. The error between the fresh and rotten classes is crucial since it influences the effectiveness of anomaly detection. The error between the rotten classes is not very important for anomaly detection as it does not necessarily decrease binary anomaly recall. Therefore, the confusion matrix gives information that cannot be obtained through classification accuracy.

C. False Negative and Error Analysis

False Negative Analysis was done in order to study the cases where images tagged as rotten fruits were falsely predicted as fresh fruits. This was especially necessary since the main purpose of this project was to detect anomalies on the surface of the fruits. As part of anomaly evaluation, the three classes of rotten fruits were considered the positive class, whereas the three classes of fresh fruits were taken as the negative class. The custom CNN made 62 false negative predictions, leading to an anomaly recall of around 95.96%. EfficientNetB0 made three false negative predictions and had an anomaly recall of about 99.80%. Thus, the decrease in the number of false negatives shows that EfficientNetB0 was better able to detect the rotten labels in the dataset. But the errors still show that having high classification accuracy does not necessarily mean that no anomalies will be missed.

$$\text{Anomaly Recall} = \frac{TP_{\text{rotten}}}{TP_{\text{rotten}} + FN_{\text{rotten}}} \tag{5}$$

In Equation 5, TP_{rotten} represents the number of images labelled as rotten that were predicted as any rotten category, while FN_{rotten} represents rotten images incorrectly predicted as fresh.

It is meant to show some sample cases of false negatives produced by the models during evaluation. This gives a chance to analyze what visual aspects of the images could have led to their misclassification into fresh fruit categories. These can include minor surface degradation, similarity in color, and other types of differences in the image's appearance. However, these should only be considered possible reasons unless proven by an analysis of the misclassified images. The difference in false negatives between the models indicates that pretraining on visual features was helpful in classifying the labeled anomalies. Still, the evaluation was conducted using images from a particular fruit dataset available in the public domain. Therefore, the findings cannot prove that the models would demonstrate equal performance in case of different lighting, backgrounds, or conditions of image acquisition. Finally, the models classify visible image labels and should not be considered food safety certification systems.

D. Error Analysis

1) Custom CNN Classification Errors: Classification errors made by the custom CNN were analyzed to understand the weaknesses of the baseline model to distinguish fresh and rotten fruit images. In particular, the model was able to reach 89.33% accuracy on the test set, which means that a part of the test images were misclassified. More specifically, there were 62 misclassified images, marked as rotten fruit that were incorrectly classified as a fresh fruit class. It is especially important to note such a classification error because it shows the inability to distinguish visual deterioration from the data. The anomaly recall of the model of 95.96% shows that most of the anomalous instances of rotten fruits were recognized by the model. Some reasons for such classification errors might lie in the similarity of color, texture, and appearance of fresh and rotten fruits. However, this assumption needs to be validated using the examples of the images misclassified by the model.

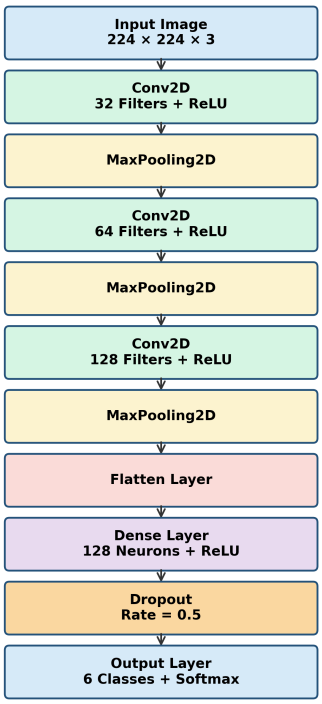


Fig. 5. Architecture of the custom CNN developed for six-class fruit surface anomaly classification.

It provides examples of misclassification of the custom CNN on the test data. Analysis of the images that the custom CNN could not classify properly makes it possible to explore the link between visual characteristics of the fruit and classification mistakes. In some cases, certain image features might look similar to features characteristic for another type of fruit or its freshness level. In other cases, the light condition, surface texture, and degree of deterioration might become a factor that complicates classification. It is impossible to state the impact of these features without analyzing the images. Mistakes show that not all visual features were equally easy for the baseline neural network model to distinguish. False negatives are particularly important since classifications of rotten fruits as fresh affect anomaly detection results negatively.

2) EfficientNetB0 False-Negative Analysis: False negative predictions made by the EfficientNetB0 were considered to understand the reasons behind the classification errors even though the network is highly efficient. EfficientNetB0 demon-

strated test accuracy of 99.89% and anomaly recall rate of about 99.80% when applied to the analyzed dataset. There were three images out of 1,534 testing images labeled as rotten that were predicted to be fresh fruit images. These examples are considered to be the cases when the model did not recognize the anomaly condition denoted in the labels. In comparison with the custom CNN, there were 62 false negatives in predictions made by the custom CNN and three false negatives made by EfficientNetB0. It means that the usage of pretrained visual features helped to improve recognition accuracy for fresh and rotten fruit images. Nonetheless, the existing false negatives show that even highly accurate classification model can make errors while predicting the images with anomalies.

It illustrates the three false negatives produced by the EfficientNetB0 model while testing. These images serve as a starting point for analyzing the visual features linked to the remaining classification mistakes of the model. The potential reasons behind the misclassification can be limited visible decay, similarity in the appearance of the fresh and rotten surfaces, or differences in the images. Still, these should be viewed as possible reasons, but not as proven, until the images are analyzed separately. The low number of false negatives demonstrates good anomaly detection capabilities of the model on the chosen testing set. Yet, the results do not prove that the model will work as well on images taken in other environments. Moreover, the classification system recognizes visual features associated with the dataset labels, but does not measure microbial contamination or food safety directly.

VII. Conclusion

The study focused on the efficiency of deep learning methods for visible surface anomaly classification in fruit images through the use of a custom-made convolutional neural network (CNN) and an ImageNet-pretrained EfficientNetB0 architecture. These tests have been done using a publicly available dataset consisting of fresh and rotten images of apples, bananas, and oranges, categorized into six classes. Classification accuracy, macro-averaged F1-score, anomaly recall, and false-negative analysis have been used as metrics to measure the capacity of both models to classify fresh and visually deteriorated fruits. The custom CNN provided an accuracy of 89.33%, a macro F1-score of 89.43%, and an anomaly recall of 95.96% that shows that it is able to learn useful visual features from the training set. On the other hand, EfficientNetB0 provided an accuracy of 99.89%, a macro F1-score of 99.90%, and an anomaly recall of 99.80%. Additionally, the instances of erroneous classification of rotten images as fresh reduced from 62 in the custom CNN case to just three with EfficientNetB0. The above observations indicate that transfer learning can aid in enhancing visual anomaly detection through the use of pre-learned features rather than solely learned features from a smaller dataset. Nevertheless, the study has limitations with its evaluations being based on three types of fruits and one publicly accessible dataset. It is possible to extend future studies to consider other food types, use other data sources, employ better data augmentation methods, and use approaches to Explainable Artificial Intelligence, such as Grad-CAM for visual explanations [12]. Future evaluations could also consider broader anomaly-detection datasets such as PKU-GoodsAD [21]. In summary, the paper demonstrates the possible benefits of using transfer learning for visual food anomalies on food surfaces while noting that the predictions based on image classification alone do not guarantee food safety.

AI-Use Declaration

I acknowledge the use of ChatGPT as an AI-assisted tool during the preparation of this research project. AI assistance was used for code debugging, language refinement, structuring and drafting parts of the research report, and preparing presentation materials. All AI-generated suggestions and content were reviewed and edited before inclusion. The experimental results were obtained through the implemented machine learning models and evaluated using the project dataset. I take full responsibility for the accuracy, interpretation, originality, and academic integrity of the submitted work.

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